

Examination Information Page

Written examination

General information about the exam:

Subject code: AI3000R

Subject name: AI for Business Applications

Examination date: 12.12.2025

Examination time: 09:00 – 13:00

Total hours: 4 hours

Responsible course manager: Sinuo Wu

Campus:
Ringerike

Faculty: USN School of Business

No. of assignments: 10 questions

No. of attachments: 0

No. of pages incl. information page and attachments: 11

Permitted aids (cf. course plan):

None

Information regarding attachments:

None

Comments:

• A: 89–100 pts • B: 77–88 pts • C: 65–76 pts • D: 53–64 pts • E: 41–52 pts • F: 0–40 pts

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Spreadsheets ☐

Line sheets ☐

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Exam assignment:

Question 1: Artificial Intelligence (10 pts)

- A) In your own words, describe the Turing Test and explain how it works. (4 pts) *(Keep your answer within 150 words.)*
- B) Provide an example of a failed application of Artificial Intelligence (AI) in the business domain. Discuss the reasons behind its failure and propose a possible solution to address the issue. (6 pts) *(Keep your answer within 200 words.)*

Question 2: Machine Learning Pipelines and Feature Selection (10 pts)

- A) Case Study

Case Description: This case presents features related to employee information within an organization, including Age, Job Status (Currently Employed/Left), Distance from Home, Employee Number, Environment Satisfaction, Gender, Job Involvement, Job Satisfaction, Marital Status, and Performance Rating. The objective is to train an AI model to predict whether an employee will leave the company and identify the key factors contributing to employee turnover.

Below is a portion of the dataset. Please read and respond to the following questions. *(Keep your answer within 100 words for each sub-question.)*

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Age	Job Status	Distance From Home	Employee Number	Environment Satisfaction	Gender	Job Involvement	Job Satisfaction	Marital Status	Performance Rating
41	Currently employed	1	1	2	Female	3	4	Single	3
49	Currently employed	8	2	3	Male	2	2	Married	4
37	Currently employed	2	4	4	Male	2	3	Single	3
33	Currently employed	3	5	4	Female	3	3	Married	3
27	Currently employed	2	7	1	Male	3	2	Married	3
32	Left	2	8	4	Male	3	4	Single	3
59	Currently employed	3	10	3	Female	4	1	Married	4
30	Left	24	11	4	Male	3	3	Divorced	4
38	Currently employed	23	12	4	Male	2	3	Single	4
36	Currently employed	27	13	3	Male	3	3	Married	3
35	Currently employed	16	14	1	Male	4	2	Married	3
29	Currently employed	15	15	4	Female	2	3	Single	3
31	Currently employed	26	16	1	Male	3	3	Divorced	3
...	...	...	...	...	...	...	...	...	...

1. Identify and list the features you would select for building the model. Also, point out the target value. (2 pts)
2. Describe the difference between feature selection and feature engineering. Additionally, create at least one new feature using feature engineering based on your knowledge. (2 pts)
3. Choose a categorical feature from the dataset that requires transformation. Describe one method to transform this feature. (2 pts)
4. Identify and list the features that need to be scaled. Suggest one method for scaling these features and explain why scaling is important for the model. (4 pts)

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Question 3: Supervised Learning (10 pts)

- A) In your own words, explain what a confusion matrix is. Additionally, describe a scenario where a confusion matrix can be applied. (4 pts) *(Keep your answer within 150 words.)*

B) Case Study

Case Description: The code below is used to build an AI model that predicts the most likely place a person will check in to. The dataset represents an artificial world with over 100,000 places located within a 10 km by 10 km square. For a given set of coordinates, the task is to return a ranked list of the most likely places. The dataset includes the following features:

- **row_id:** ID of the check-in event
- **x, y:** Coordinates of the check-in
- **accuracy:** Location accuracy
- **time:** Timestamp of the check-in
- **place_id:** ID of the business (this is the target variable the model is predicting)

Please review the provided code and respond to the following questions. *(Keep your answer within 100 words for each sub-question.)*

```

1  import pandas as pd
2  from sklearn.model_selection import train_test_split
3  from sklearn.preprocessing import StandardScaler
4  from sklearn.neighbors import KNeighborsClassifier
5  from sklearn.metrics import accuracy_score
6
7  data = pd.read_csv(r'C:\Users\ws\Desktop\FB competition.csv')
8
9  time_value = pd.to_datetime(data["time"], unit="s")
10 data["day"] = time_value.dt.day
11 data["weekday"] = time_value.dt.weekday
12 data["hour"] = time_value.dt.hour
13
14 X = data[["x", "y", "accuracy", "day", "weekday", "hour"]]
15 y = data["place_id"]
16
17 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
18
19 scaler = StandardScaler()
20 X_train_scaled = scaler.fit_transform(X_train)
21 X_test_scaled = scaler.transform(X_test)
22
23 k = 3
24 knn = KNeighborsClassifier(n_neighbors=k)
25 knn.fit(X_train_scaled, y_train)
26
27 y_pred = knn.predict(X_test_scaled)
28
29 print("Predicted labels:")
30 print(y_pred)
31 accuracy = accuracy_score(y_test, y_pred)
32 print("Accuracy:", accuracy)

```

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1. Explain how the K-Nearest Neighbor algorithm works and what is meant by $K = 3$. (4 pts)
2. Describe the evaluation method used for the model and explain how it works. (2 pts)

Question 4: Unsupervised Learning (10 pts)

A) In your own words, explain the difference between supervised learning and unsupervised learning. (4 pts) *(Keep your answer within 100 words.)*

B) Case Study

Case Description: This case involves customer information from a store, including the following features: Customer ID, Gender, Age, Location, Annual Spending, Purchase Frequency, and their Preferred Product Category. The objective is to train an AI model to group customers into different segments, enabling the store to create customized advertising strategies. Below is a portion of the dataset. Please review it and respond to the following questions. *(Keep your answer within 150 words for each sub-question.)*

Customer ID	Age	Gender	Location	Annual Spending (USD)	Purchase Frequency	Preferred Product Category
C001	24	Female	Oslo	850	6	Beauty
C002	37	Male	Bergen	1950	12	Electronics
C003	29	Female	Trondheim	1200	9	Clothing
C005	33	Female	Oslo	680	4	Beauty
C006	52	Male	Tromsø	2800	10	Electronics
C007	21	Female	Oslo	320	3	Clothing
C008	40	Male	Bergen	1450	7	Beauty

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C009	27	Female	Drammen	2100	12	Electronics
C010	31	Male	Oslo	900	5	Clothing
...	...	...	...	...	...	...

1. Select the features you would use for the AI model. List the selected features and describe the model-building process, including steps from data preparation to model evaluation. (4 pts)

2. In your own words, explain how the clusters (segments) created by your AI model could be utilized effectively. (2 pts)

Question 5: Natural Language Processing (10 pts)

A) In your own words, describe the complete NLP pipeline that can be applied to the following sentence. Explain what happens at each step:

“The goal of natural language processing is to develop algorithms that allow computers to interpret unstructured language and help them move closer to genuine understanding.”
(4 pts) (Keep your answer within 150 words.)

B) Case study.

Case Description: This case study involves the use of Natural Language Processing (NLP) techniques for sentiment analysis. The objective is to determine the emotional tone

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conveyed in a body of text. The dataset for this analysis consists of customer reviews collected from a restaurant's online platforms. Each review contains textual data reflecting customers' opinions about the food and service, along with a corresponding rating score (ranging from 1 to 5).

The goal is to train an AI model to perform sentiment analysis using this dataset. Below is a portion of the dataset. Please read it and respond to the following questions. (Keep your answer within 150 words for each sub-question.)

Review Text	Rating
Had B1G1 dinner. Food was average. But a lady staff came to us and took my phone to write a review of her own. I was stunned. She wrote "best beef teheri in town" but I didn't even try teheri. Bunch of SCAMMERS.	2
Totally disappointed by visiting at restaurant. Not much option for vegetarian customers. Poor & excessive billing. Its waste of money for vegetarian guest. Poor Billing. In spite sharing elite membership & status they didn't offer appropriate discount.	1
Seasonal tastes offers a great buffet, with Western, Indian and other Asian options, varying by the time of day. There is a wide selection of choices and quality is good. The restaurant is located in the Westin, and the location implies good safety.	4
The Seasonal Taste mainly serves buffet lunch & dinner and have the option to choose from the several mouthwatering items! Chef prepares Soup, Salad, Pastas, Meat, Prawns etc. as per your taste. In general, taste of the foods are excellent.	3
The food spread is great. I was looking for vegetarian local fare which I got. Amazingly cooked. Stunning options. Dessert options are very good. Enjoyed the food.	4
Service has been top notch over the years. The food is exceptional. The rooms are clean. Very excellent service. Thanks	4
Seasonal tastes restaurant nice view. but service was not good. and I came my family purpose. but food quality very bad. i will not come in this place	1
I came my friend purfuses the seasonal tastes restaurant. food quality is not well. service is good. nice view. but especially fasta quality disgusting so i decided that next time i will not come here Again	2
I've been to so many countries and restaurants, and the breakfast and dinner at Seasonal Tastes is the best I've seen! Many delicious options and the great staff make the experience unforgettable! I can't not forget to mention Mizan for the absolutely impeccable service.	1
When the Westin opened this restaurant was fabulous. Many salads contained imported ingredients, desserts included stuffed fresh figs and elaborate pastry for 2500 taka per person (\$29). The price has skyrocketed to almost 5000 taka per person (\$59) and the food is terrible.	2
It was a great experience considering both the food and ambience. Top notch service. Hope to come again... Thank you.	5
Very good service, friendly & helpful staffs. Good was great! Family loved the experience. Would love to visit again.	5
The food is really fresh and good.....the ambience is really nice...the staffs are also very friendly..definitely a 5 start	5
Very gentle environment, good food, relaxing Interior space. Everyone service was excellent. Staff are very polite	4
.....	...

1. Create sentiment labels to categorize the "Rating" into three classes. (2 pts)

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2. Recommend a suitable word vectorizer to transform the text into numerical vectors and suggest an appropriate algorithm to build the NLP model. (4 pts)

Question 6: Computer Vision (10 pts)

- A) In your own words, explain what "features" in an image are. Additionally, describe how a color is represented using the RGB model. (4 pts) *(Keep your answer within 150 words.)*

- B) Case study.

Case Description. This case study involves using computer vision techniques for image classification. The objective is to classify images into two categories: dog or cat. The dataset consists of images collected from online platforms, including 100 images of dogs and 500 images of cats. The goal is to train an AI model for image classification using this dataset. *(Keep your answer within 150 words for each sub-question.)*

1. Point out at least one potential issue with this dataset for training the AI model. (2 pts)

2. Suggest an approach, using Python, that can be applied to increase the number of training samples. Briefly explain how this approach works. (4 pts)

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Question 7: Neural network (10 pts)

A) Case Study

Case Description: This case involves building a neural network for image classification, where the objective is to classify images into three categories: triangle, square, and circle. The goal is to design and train a neural network for this classification task. *(Keep your answer within 150 words for each sub-question.)*

1. In your own words, describe the forward propagation process in this neural network. For example, start with a circle image and explain how the image is processed step-by-step within the neural network. (4 pts)

2. In your own words, explain the purpose of the backpropagation process in this neural network. Additionally, describe what happens during this process. (6 pts)

Question 8: Deep Learning (10 pts)

A) In your own words, explain the different types of layers in Convolutional Neural Networks (CNNs) and their roles in the network. (4 pts) *(Keep your answer within 150 words.)*

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B) Case study.

Case Description: This case involves training a Convolutional Neural Network (CNN) to classify images into three categories: sunflower, rose, or dandelion. The figure below illustrates the model's training progress over 18 epochs, showing the Accuracy and Loss for both the training and validation datasets. *(Keep your answer within 150 words for each sub-question.)*



1. In your own words, describe the following terms in the context of training a deep learning model: Epoch, Accuracy, and Loss. (2 pts)

2. The figure suggests a potential model overfitting issue. Describe the specific evidence in the figure that supports this observation and suggest one approach to mitigate it. (3 pts)

Question 9: Chatbot (10 pts)

A) In your own words, explain the difference between rule-based chatbots and AI-powered chatbots. Provide at least one example of each type. (6 pts) *(Keep your answer within 150 words.)*

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- B) Describe how AI-powered chatbots can specifically benefit programmers. (4 pts) *(Keep your answer within 150 words.)*

Question 10: Philosophy, ethics, and safety of AI (10 pts)

- A) Advanced human-like AI assistants, designed to look and behave like real people just like in sci-fi movies, might roam the streets in the future, performing tasks, interacting socially, and even displaying emotions. Supporters argue that such AI could improve daily life, assist the elderly, and enhance productivity. Critics worry that human-like AI might blur the line between humans and machines, reduce genuine social interaction, or create ethical challenges. Do you think society should embrace human-like AI assistants in public life? Please explain your viewpoint and reasoning. (5 pts) *(Within 150 words)*
- B) Artificial intelligence tools, including generative AI, self-driving vehicles, and various robots, are becoming an integral part of daily life. Some argue that AI can enhance human intelligence by providing instant answers to nearly all questions, supporting learning, and improving decision-making. Others worry that reliance on AI may reduce critical thinking, creativity, and problem-solving skills, especially when the answers provided by AI are not completely accurate or ethically sound. Do you think AI is making humans smarter or dumber? Please explain your viewpoint and reasoning. (5 pts) *(Within 150 words)*

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